Nuisance-Prompt Tuning: Improving Few-Shot Out-of-Distribution Detection via Adaptive Background Modeling

Anonymous Author(s)
Affiliation
Address
email

Abstract

Deploying machine learning models safely requires detecting when inputs differ from training data—a challenge that becomes critical when only limited labeled examples are available. We present Nuisance-Prompt Tuning (NPT), a novel approach for few-shot out-of-distribution detection that explicitly models background patterns through a learnable nuisance prompt and dynamically weighted background modeling. Unlike existing methods such as LoCoOp (AUROC: 90.9%, FPR95: 42.0%) that rely on heuristic patch regularization, NPT introduces a dedicated nuisance prompt to capture background features, combined with attention-weighted patch supervision and margin-based repulsion for robust class-background separation. Our adaptive scheduling strategy uses cosine annealing to emphasize background modeling early in training through high loss weights before gradually transitioning to class-specific refinement, implementing a curriculum learning approach that prevents interference between competing objectives. On standard benchmarks (iNaturalist, SUN, Places365, Texture), NPT achieves a 25% relative FPR95 reduction and improves overall AUROC from 90.9% to 93.0% (FPR95: 31.5%). The learnable nuisance prompt provides superior explicit background modeling compared to implicit regularization approaches, demonstrating that systematically modeling what we don't want to detect can be more powerful than implicitly regularizing against it.

1 Introduction

Consider a medical imaging system trained on limited chest X-ray data that must reliably detect when presented with MRI scans or other out-of-distribution inputs. Similarly, an autonomous vehicle's perception system trained on limited urban driving data must detect novel scenarios like off-road terrain or unusual weather conditions. These scenarios exemplify the critical challenge of few-shot out-of-distribution (OOD) detection—identifying when test inputs differ from the training distribution when only minimal labeled data is available (Hendrycks & Gimpel, 2017; Lu et al., 2024a). Such capability is fundamental to deploying machine learning systems safely in real-world scenarios (Yang et al., 2021, 2022).

Traditional OOD detection methods require extensive training data or complex architectural modifications (Liang et al., 2018; Lee et al., 2018; Liu et al., 2020; Huang et al., 2021), making them impractical for few-shot settings. Recent advances in vision-language models, particularly CLIP (Radford et al., 2021), have opened new avenues through prompt learning approaches such as CoOp (Zhou et al., 2022a) and related methods (Li et al., 2022). These methods leverage pre-trained vision-language representations to learn task-specific prompts from minimal data, but standard approaches like CoOp tend to overfit to background features in ID images (Chen et al., 2025).

Citation: [Chen+, 2025] is not related in this context.

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LoCoOp (Miyai et al., 2023a) addresses background overfitting by introducing local regularization through entropy maximization on ID-irrelevant patches. However, LoCoOp has three fundamental limitations that constrain its effectiveness: it relies on heuristic top-K ranking to identify irrelevant patches, which can be unstable across training; it uses fixed hyperparameters throughout training, preventing adaptive emphasis on different learning phases; and it lacks explicit modeling of background patterns, instead relying on implicit regularization. These constraints motivate a paradigm shift toward more principled background modeling approaches that systematically capture nuisance information.

We propose **Nuisance-Prompt Tuning (NPT)**, which addresses these limitations through explicit background modeling and adaptive training strategies. Our key insight is that effective few-shot OOD detection requires systematically modeling what constitutes background or nuisance information, rather than relying on implicit regularization. NPT introduces a learnable nuisance prompt that captures background patterns, complemented by attention-weighted patch supervision and adaptive loss scheduling.

NPT incorporates four key innovations: (1) **Explicit nuisance modeling** through a dedicated learnable prompt that systematically captures background patterns; (2) **Attention-weighted patch supervision** that uses CLIP's attention mechanisms to identify background regions without heuristic thresholding (Leem & Seo, 2024; Guo et al., 2023); (3) **Margin-based repulsion** that ensures robust separation between class and nuisance representations in embedding space (Deng et al., 2018; Gupta et al., 2023); and (4) **Adaptive loss weight scheduling** that emphasizes background modeling early before transitioning to class-specific refinement (Bengio et al., 2009; Gong et al., 2019).

Citation: [Gupta+, 2023] is not related in this context.

We evaluate NPT on standard benchmarks including iNaturalist (Van Horn et al., 2018), SUN (Xiao et al., 2010), Places365 (Zhou et al., 2017), and Texture (Cimpoi et al., 2014) as OOD datasets with ImageNet (Deng et al., 2009) as in-distribution data. NPT achieves significant improvements over LoCoOp: 93.0% overall AUROC (vs. 90.9%) and 25% relative FPR95 reduction (31.5% vs. 42.0%). Comprehensive ablation studies validate each component's importance and reveal insights into effective background modeling strategies.

Our contributions demonstrate that explicit background modeling fundamentally changes the approach to few-shot OOD detection, providing a paradigm shift from implicit regularization to systematic nuisance modeling with practical improvements for real-world deployment.

2 Related Work

2.1 Traditional OOD Detection

Traditional OOD detection methods include confidence-based approaches using Maximum Softmax Probability (Hendrycks & Gimpel, 2017) and temperature scaling methods like ODIN (Liang et al., 2018; Guo et al., 2017; Manna et al., 2023), distance-based approaches through Mahalanobis distance (Lee et al., 2018), and energy-based methods (Liu et al., 2020). Recent advances include gradient-based detection (Huang et al., 2021; Sharifi et al., 2024), virtual outlier synthesis (Du et al., 2022; Kalina, 2025), feature-based methods like ViM (Wang et al., 2022), and ensemble approaches (Lakshminarayanan et al., 2017). Proto-OOD (Chen et al., 2024) enhanced OOD object detection through prototype feature similarity. Unlike NPT, these methods typically require extensive training data and struggle in few-shot scenarios.

Citation: [Guo+, 2017] and [Manna+, 2023] are not related in this context.

Citation: [Kalina+, 2025] is not related in this context.

2.2 Vision-Language Models for Few-Shot Learning

CLIP (Radford et al., 2021) transformed few-shot learning through learnable prompt optimization (Li et al., 2022). CoOp (Zhou et al., 2022a) pioneered context optimization learning continuous context vectors (Xing et al., 2022), while CoCoOp (Zhou et al., 2022b) extended this with conditional prompts. Alternative approaches include Tip-Adapter (Zhang et al., 2022) for training-free adaptation (Farhadzadeh et al., 2025), visual prompt tuning (Jia et al., 2022; Wangni, 2024), and prefix tuning (Li & Liang, 2021; Yang & Liu, 2022). Unlike these classification-focused methods, NPT explicitly addresses OOD detection through systematic background modeling.

2.3 CLIP-based OOD Detection

CLIP has enabled new OOD detection approaches through vision-language representations (Lu et al., 2024b). Early work explored zero-shot detection using CLIP features (Esmaeilpour et al., 2022; Fort et al., 2021; Atigh et al., 2025), while MCM (Ming et al., 2022) and GL-MCM (Miyai et al., 2023b) developed sophisticated scoring functions (Peng et al., 2024). However, most methods focus on zero-shot settings rather than few-shot adaptation with explicit background modeling.

Citation: [Atigh+, 2025] is not related in this context.

Citation: [Peng+, 2024] is not related in this context.

2.4 Background and Nuisance Modeling

Explicit modeling of background information has been explored across vision tasks. Attention mechanisms identify task-relevant regions (Vaswani et al., 2017; Dosovitskiy et al., 2021; Leem & Seo, 2024; Guo et al., 2023), while outlier exposure (Hendrycks et al., 2019) demonstrates the importance of negative sample modeling. Texture bias research (Geirhos et al., 2018) highlights background overfitting challenges in ImageNet-trained models. Unlike these approaches that implicitly handle background, NPT introduces explicit nuisance prompt learning.

2.5 Curriculum Learning and Adaptive Training

Curriculum learning (Bengio et al., 2009) shows that organizing training complexity improves optimization. Adaptive training strategies include dynamic loss weighting (Gong et al., 2019; Zhao et al., 2015; Luo et al., 2021) and learning rate scheduling (Subramanian & Ganapathiraman, 2023; Singh et al., 2025). NPT incorporates these principles through adaptive loss weight scheduling that treats background modeling as a curriculum problem.

Unlike existing approaches that rely on heuristic regularization or implicit background handling, NPT introduces principled explicit nuisance modeling through a dedicated learnable prompt combined with adaptive training strategies, providing a fundamental shift from implicit to explicit background modeling for robust few-shot OOD detection.

3 Method

3.1 Overview

We tackle few-shot out-of-distribution (OOD) detection using vision-language models, where only a few labeled in-distribution (ID) samples are available for training. Our work builds upon LoCoOp (Miyai et al., 2023a), a local regularized context optimization method that performs OOD detection via prompt learning with CLIP (Radford et al., 2021).

3.2 Preview of Baseline Method

The baseline LoCoOp method addresses limitations of standard prompt learning approaches like CoOp (Zhou et al., 2022a) for OOD detection. While CoOp brings ID images closer to their corresponding class text embeddings, it inadvertently also brings text embeddings closer to ID-irrelevant features (backgrounds, objects) in ID images. This leads to high confidence scores for OOD images containing similar irrelevant features.

LoCoOp addresses this by identifying ID-irrelevant regions in local CLIP features and treating them as pseudo-OOD features during training. Specifically, it:

1. Extracts local features from CLIP's vision transformer using value projections from visual to textual space
2. Identifies ID-irrelevant regions where the ground truth class does not appear in top-K predictions
3. Applies entropy maximization on these regions to push them away from all ID class text embeddings

The LoCoOp objective combines standard prompt learning loss with OOD regularization:

$$\mathcal{L}_{LoCoOp} = \mathcal{L}_{global} + \lambda_{entropy}\mathcal{L}_{entropy} \tag{1}$$